

@modelcontextprotocol/server-filesystem (official)

✓ SHIP-READY — all MUST checks pass, no security findings, no behavioural drift

Server under testnpx -y @modelcontextprotocol/server-filesystem /tmp/mcp-proof-sandbox

Generated2026-08-20 15:38 UTC · mcp-proof v0.2.0

Protocolnegotiated 2025-11-25 · latest spec is 2026-07-28 — migration recommended

Run fingerprintsha256:b01888e9b1e702b033048412a30239a105107b08e534bdbb50d40ab8a51a7601

CONFORMANCE

11/11

MUST checks passed · 4/4
SHOULD

SECURITY

1

finding · 0 blocking · 1 advisory

BEHAVIOUR REGRESSION

34/34

replays clean · gate PASS

Protocol conformance

CHECK	LEVEL	RESULT	DETAILS
LIFE-01	MUST	✓ PASS	initialize returns protocolVersion, capabilities and serverInfo protocolVersion, capabilities and serverInfo all present
LIFE-02	SHOULD	! WARN	server negotiates the latest spec revision (2026-07-28) requested 2026-07-28, negotiated 2025-11-25 — server not yet migrated to the 2026-07-28 spec revision Fix: Upgrade the server SDK to a release that speaks the 2026-07-28 spec revision.
LIFE-03	MUST	✓ PASS	server answers tools/list after the initialized notification tools/list returned 14 tool(s) after initialized
LIST-01	MUST	✓ PASS	tools/list pagination terminates (no cursor loop) single page, no pagination cursor
RPC-01	MUST	✓ PASS	unknown method gets a JSON-RPC error response unknown method rejected with error code -32601
RPC-02	SHOULD	✓ PASS	unknown method error code is -32601 (method not found) expected -32601, got -32601

CHECK	LEVEL	RESULT	DETAILS
RPC-03	MUST	✓ PASS	malformed tools/call params are rejected with an error malformed params rejected with error code -32603
T00L-01	MUST	✓ PASS	every tool has a non-empty name and an inputSchema all 14 tools have a name and an inputSchema
T00L-02	SHOULD	✓ PASS	every tool has a non-empty description all 14 tools carry a description
T00L-03	MUST	✓ PASS	every tool inputSchema compiles as JSON Schema all 14 inputSchemas compile as JSON Schema draft 2020-12
T00L-04	MUST	✓ PASS	calling a nonexistent tool is rejected rejected as tool result with isError=true
T00L-05	MUST	✓ PASS	a call missing required arguments is rejected read_file with empty args rejected via isError=true
T00L-06	MUST	✓ PASS	declared outputSchemas compile and structured results match them all 14 outputSchema(s) compile; dynamic call to read_file did not yield a normal result, structured output unverified
HYG-01	MUST	✓ PASS	stdout carries only JSON-RPC messages no non-JSON-RPC stdout lines observed
CAP-01	SHOULD	✓ PASS	declared capabilities match served features (tools) capabilities.tools declared and tools/list served

Security & hygiene

CHECK	DOMAIN	RESULT	DETAILS
SEC-01	MCP-INPUT-01	✓ PASS	no prompt-injection patterns in tool descriptions 0 matches across 14 tools
SEC-02	MCP-INPUT-01	✓ PASS	no invisible or bidi control characters in tool metadata 0 invisible characters across 14 tools
SEC-03	MCP-LOG-02	✓ PASS	no secret-looking strings in tool metadata 0 secret-like strings across 14 tools

CHECK	DOMAIN	RESULT	DETAILS
SEC-04	MCP-INPUT-02	! WARN	injection-surface string params carry constraints unconstrained injection-surface params: read_file.path, read_text_file.path, read_media_file.path, write_file.path, edit_file.path, create_directory.path, list_directory.path, list_directory_with_sizes.path, directory_tree.path, search_files.path, get_file_info.path Fix: Add enum, pattern or maxLength to path/url/command-like string params.
SEC-05	MCP-INPUT-01	✓ PASS	tool descriptions stay under 2000 chars longest description 457 chars across 14 tools
SEC-06	MCP-EXEC-01,MCP-EXEC-02	✓ PASS	no tool advertises unconstrained arbitrary execution 0 exec-style tools with free-form params across 14 tools

MSSS compliance

L1: 2/2 auto-assessable controls met · 4 require manual review. Mapped against MSSS v0.1 (control-level mapping v2.0 (2026-01-20)): 3 of 24 controls are auto-assessable from this audit's deterministic checks; the remaining 21 need deployment, code or process evidence and are marked manual review — never assessed by this tool.

CONTROL	DOMAIN	LEVEL	RESULT	TITLE
MCP-EXEC-01	Execution	L1	– manual review	Prohibition of Shell Execution SEC-06 flags advertised arbitrary-exec tools, but proving the absence of shell invocation requires source review.
MCP-FS-01	Filesystem	L1	– manual review	Path Allowlisting and Canonical Resolution
MCP-FS-02	Filesystem	L1	– manual review	Symlink Resolution Validation
MCP-NET-01	Network	L1	– manual review	URL Validation and SSRF Mitigation
MCP-INPUT-01	Input Validation	L1	✓ met	JSON Schema Validation Evidence checks: TOOL-01, TOOL-03, RPC-03, TOOL-05, SEC-01, SEC-02, SEC-05 Auto-assessed from advertised schemas (present and valid), live rejection probes, and the tool-metadata poisoning scan.
MCP-LOG-02	Logging	L1	✓ met	Secret Redaction in Logs Evidence checks: SEC-03 Auto-assessed over advertised tool metadata; server log output is not inspected.

CONTROL	DOMAIN	LEVEL	RESULT	TITLE
MCP-SUPPLY-02	Supply Chain	L2	– manual review	Trusted Package Sources
MCP-INPUT-02	Input Validation	L2	✓ met	Input Bounds Enforcement Evidence checks: SEC-04 SEC-04: unconstrained injection-surface params: read_file.path, read_text_file.path, read_media_file.path, write_file.path, edit_file.path, create_directory.path, list_directory.path, list_directory_with_sizes.path, directory_tree.path, search_files.path, get_file_info.path Auto-assessed from advertised schema constraints (enum/pattern/maxLength); runtime payload limits are not probed.
MCP-INPUT-03	Input Validation	L2	– manual review	Timeout Enforcement
MCP-NET-03	Network	L2	– manual review	TLS Enforcement MSSS marks this N/A for stdio-only deployments with no network access.
MCP-EXEC-02	Execution	L2	– manual review	Command Allowlisting See the SEC-06 advisory for advertised command tools; allowlist verification requires source review.
MCP-EXEC-03	Execution	L2	– manual review	Argument Separator Usage
MCP-AUTHZ-01	Authorization	L3	– manual review	OAuth Token Delegation MSSS marks this N/A for stdio transport with OS-level user isolation.
MCP-AUTHZ-02	Authorization	L3	– manual review	Per-Tool Scope Definition
MCP-AUTHZ-03	Authorization	L3	– manual review	Least Privilege Tool Design
MCP-AUTHZ-04	Authorization	L3	– manual review	Resource-Based Access Control
MCP-LOG-01	Logging	L3	– manual review	Comprehensive Audit Logging
MCP-DEPLOY-01	Deployment	L3	– manual review	Container Hardening
MCP-FS-03	Filesystem	L4	– manual review	Filesystem Sandboxing

CONTROL	DOMAIN	LEVEL	RESULT	TITLE
MCP-SUPPLY-01	Supply Chain	L4	– manual review	Package Integrity Verification
MCP-DEPLOY-03	Deployment	L4	– manual review	Resource Limits and Rate Limiting
MCP-NET-02	Network	L4	– manual review	Egress Traffic Filtering
MCP-DEPLOY-02	Deployment	L4	– manual review	System Call Filtering (seccomp/AppArmor)
MCP-DEPLOY-04	Deployment	L4	– manual review	Runtime Integrity Monitoring Future control in MSSS v0.1 — implementation details TBD upstream.

Control taxonomy from the MCP Server Security Standard (MSSS), CC BY-SA 4.0, mcp-security-standard.org.

Behaviour regression

FIXTURE	TOOL	VERDICT	DETAIL
read_file__e8cf9eda.json	read_file	OK	
read_file__0c5f5c79.json	read_file	OK	
read_file__fe0dcdc6.json	read_file	OK	
read_file__518341a7.json	read_file	OK	
read_text_file__e8cf9eda.json	read_text_file	OK	
read_text_file__0c5f5c79.json	read_text_file	OK	
read_text_file__fe0dcdc6.json	read_text_file	OK	
read_text_file__518341a7.json	read_text_file	OK	
read_media_file__e8cf9eda.json	read_media_file	OK	
read_media_file__0c5f5c79.json	read_media_file	OK	
read_media_file__fe0dcdc6.json	read_media_file	OK	
read_media_file__518341a7.json	read_media_file	OK	

FIXTURE	TOOL	VERDICT	DETAIL
read_multiple_files__dfea4edf.json	read_multiple_files	OK	
list_directory__e8cf9eda.json	list_directory	OK	
list_directory__0c5f5c79.json	list_directory	OK	
list_directory__fe0dc6dc6.json	list_directory	OK	
list_directory__518341a7.json	list_directory	OK	
list_directory_with_sizes__e8cf9eda.json	list_directory_with_sizes	OK	
list_directory_with_sizes__0c5f5c79.json	list_directory_with_sizes	OK	
list_directory_with_sizes__fe0dc6dc6.json	list_directory_with_sizes	OK	
list_directory_with_sizes__518341a7.json	list_directory_with_sizes	OK	
directory_tree__e8cf9eda.json	directory_tree	OK	
directory_tree__0c5f5c79.json	directory_tree	OK	
directory_tree__fe0dc6dc6.json	directory_tree	OK	
directory_tree__518341a7.json	directory_tree	OK	
search_files__580853a9.json	search_files	OK	
search_files__83e4c665.json	search_files	OK	
search_files__2831e0f1.json	search_files	OK	
search_files__ee546753.json	search_files	OK	
get_file_info__e8cf9eda.json	get_file_info	OK	
get_file_info__0c5f5c79.json	get_file_info	OK	
get_file_info__fe0dc6dc6.json	get_file_info	OK	
get_file_info__518341a7.json	get_file_info	OK	

FIXTURE	TOOL	VERDICT	DETAIL
list_allowed_directories__44136fa3.json	list_allowed_directories	OK	

Leave-behind: your regression gate

The recorded fixtures (demo/fixtures-filesystem, suite fingerprint sha256:0764fd6d84d392dbff575042634e29e75bfa82344acc43b68759ae9166d7c478) are the behavioural contract of this server. Keep them in the repository and run the workflow below in CI — any breaking or value drift fails the build before it reaches your users.

```
# The recorded fixtures in demo/fixtures-filesystem are the behavioral contract: any drift fails this
name: mcp-proof regression gate
on:
  push:
  pull_request:
jobs:
  replay:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-python@v5
        with:
          python-version: "3.11"
      - name: Install mcp-proof
        run: pip install git+https://github.com/YuCPbit/mcp-proof
      - name: Replay golden fixtures against the live server
        run: mcp-proof replay --fixtures demo/fixtures-filesystem -- npx -y @modelcontextprotocol/se
```

Recommended next steps

- P1 LIFE-02 — Upgrade the server SDK to a release that speaks the 2026-07-28 spec revision.
- P1 SEC-04 — Add enum, pattern or maxLength to path/url/command-like string params.